

# Usability Study Report

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## Application scope

**Application description:** HabitForge is a personal habit tracking web app that lets users create daily and weekly habits, log completions, track streaks, and view analytics on their consistency over time.

**Users:** College students and young professionals aged 18-30 who want to build consistent routines such as daily coding practice, gym workouts, or reading, and need measurable feedback to stay motivated.

**Data description:** The app stores user accounts, habits (name, category, frequency), completion logs (date, notes), weekly goal targets per habit, and calculates streaks and completion percentages from the log history.

Main tasks - use cases: (specific, measurable, concrete)

- T1: You want to start tracking a new habit. Create a daily habit called "Morning Run" in the Fitness category.
- T2: You completed your Morning Run today. Log a completion for it and add a note saying, "Felt great."
- T3: You want to see how consistent you have been. Go to the Analytics page and find your current streak for Morning Run.
- T4: You want to set a weekly goal for Morning Run. Set a target of 80% weekly completion.
- T5: You no longer want to track Morning Run. Delete the habit.

# Experiment

## Preparation

### Introduction

Hi, thank you for taking the time to help me today. I'm testing a web app called HabitForge, a habit tracking tool. I want to be clear that I'm testing the app, not you. There are no wrong answers, and you can stop at any time.

I'm going to ask you to think out loud as you use the app, just narrate what you're looking at, what you're thinking, and what you're trying to do. It helps me understand your experience even when you're confused or stuck.

I'll be recording this session for research purposes only. The recording won't be shared publicly. Do you consent to being recorded?

### Demographics

Great. Before we start, I have a few quick questions about you.

1. What is your age range? (18-21 / 22-25 / 26-30 / 30+)
2. How comfortable are you with using web apps on a scale of 1-5? (1 = not comfortable, 5 = very comfortable)
3. Have you used a habit tracking app before? (Yes / No)
4. If yes, which one? (optional)
5. How often do you try to build or track personal habits? (Never / Occasionally / Regularly)

### Script of the tasks to be read to participants:

- Before I give you any specific tasks, I'd like you to open the app and just explore it for 2 minutes. Don't try to do anything specific, just look around and tell me what you think the app does and what you can do with it.

<https://habit-forge-fvl4.onrender.com> [testuser / password123]

Based on what you saw, what do you think this app is for?

- **T1 (Create a habit)** - Imagine you've decided to start going for a morning run every day and you want to track it. Please create a new habit for this.

Success criteria: Habit named anything related to running appears on the dashboard.

- **T2 (Log a completion)** - You just finished your morning run today. Please record that you completed it and add a short note about how it went.

Success criteria: A log entry appears for today's date with a note.

- **T3 (Find streak information)** - You're curious about how consistent you've been with your habits. Please find where you can see your streaks and weekly completion stats.

Success criteria: Participant navigates to Analytics and finds streak data.

- **T4 (Set a weekly goal)** - You want to challenge yourself to complete your morning run habit at least 80% of the time each week. Please set that as a goal.

Success criteria: A weekly goal of 80% is saved under the habit in Analytics.

- **T5 (Delete a habit)** - You've decided you no longer want to track your morning run habit. Please remove it from the app.

Success criteria: Habit no longer appears on the dashboard.

### Post questionnaire Likert Scales

*1 = Strongly Disagree, 5 = Strongly Agree*

- **T1 (Create a habit)**  
Creating the habit was straightforward. (1-5)  
I knew where to go to create a habit. (1-5)
- **T2 (Log a completion)**  
Logging a completion was easy. (1-5)  
I understood how to add a note to a log. (1-5)
- **T3 (Find streak information)**  
Finding my streak information was easy. (1-5)  
The Analytics page was easy to navigate. (1-5)
- **T4 (Set a weekly goal)**  
Setting a weekly goal was intuitive. (1-5)  
I understood where the goal setting feature was. (1-5)
- **T5 (Delete a habit)**  
Deleting a habit was easy. (1-5)

I felt confident the habit and its data would be removed. (1-5)

- **Overall**

The app was easy to use overall. (1-5)

I would use this app to track my own habits. (1-5)

Any comments or suggestions for improvement? (open text)

That's everything, thank you so much for your time. Your feedback is really valuable and will help improve the app. Do you have any final thoughts or anything you want to add?

## Experiment Notes

**Participant 1 - <https://youtu.be/tK15hy9jOpI>**

### Demographics answers:

1. What is your age range? (18-21 / 22-25 / 26-30 / 30+)

**Ans.** 20 years old

2. How comfortable are you with using web apps on a scale of 1-5? (1 = not comfortable, 5 = very comfortable)

**Ans.** 5

3. Have you used a habit tracking app before? (Yes / No)

**Ans.** Yes

4. If yes, which one? (optional)

**Ans.** Apple fitness

5. How often do you try to build or track personal habits? (Never / Occasionally / Regularly)

**Ans.** Occasionally

### Detailed notes:

- **Notes for initial approach:** It is an app to track our fitness
- **T1 (Create a habit):** Cleanly executed
- **T2 (Log a completion):** Cleanly executed
- **T3 (Find streak information):** Cleanly executed
- **T4 (Set a weekly goal):** Cleanly executed
- **T5 (Delete a habit):** Cleanly executed

### Post-test questionnaire results (Likert scales):

- **T1 (Create a habit): -**

Creating the habit was straightforward: 5

I knew where to go to create a habit: 5

- **T2 (Log a completion): -**

Logging a completion was easy: 4

I understood how to add a note to a log: 5

● **T3 (Find streak information): -**

Finding my streak information was easy: 5

The Analytics page was easy to navigate: 5

● **T4 (Set a weekly goal): -**

Setting a weekly goal was intuitive: 4

I understood where the goal setting feature was: 4

● **T5 (Delete a habit): -**

Deleting a habit was easy: 4

I felt confident the habit and its data would be removed: 5

**Overall**

- The app was easy to use overall: 4
- I would use this app to track my own habits: 3
- Any comments or suggestions for improvement?

[0's flattened and date range filter by 'this year' missing and 2 green tiles on analytics page]

**Participant 2 - <https://youtu.be/kXzFm8jVF4g>**

Demographics answers:

1. What is your age? (18-21 / 22-25 / 26-30 / 30+)

**Ans.** 35

2. How comfortable are you with using web apps on a scale of 1-5? (1 = not comfortable, 5 = very comfortable)

**Ans.** 3

3. Have you used a habit tracking app before? (Yes / No)

**Ans.** No

4. If yes, which one? (optional)

**Ans.** N/A

5. How often do you try to build or track personal habits? (Never / Occasionally / Regularly)

**Ans.** Occasionally

Detailed notes:

- **Notes for initial approach:** It is for the fitness tracker and the daily habit tracker app
- **T1 (Create a habit):** Cleanly executed
- **T2 (Log a completion):** Cleanly executed. Add a note to the log has scope to improve
- **T3 (Find streak information):** Cleanly Executed
- **T4 (Set a weekly goal):** Cleanly Executed
- **T5 (Delete a habit):** Cleanly Executed

Post-test questionnaire results (Likert scales):

- **T1 (Create a habit): -**

Creating the habit was straightforward: 5

I knew where to go to create a habit: 5

- **T2 (Log a completion): -**

Logging a completion was easy: 5

I understood how to add a note to a log: 4

- **T3 (Find streak information): -**

Finding my streak information was easy: 5

The Analytics page was easy to navigate: 5

- **T4 (Set a weekly goal): -**

Setting a weekly goal was intuitive: 5

I understood where the goal setting feature was: 5

- **T5 (Delete a habit): -**

Deleting a habit was easy: 5

I felt confident the habit and its data would be removed: 5

- **Overall**

The app was easy to use overall: 5

I would use this app to track my own habits: 5

Any comments or suggestions for improvement?

[Nothing to improve]

**Participant 3 - <https://youtu.be/7WKnjlzaQiE>**

Demographics answers:

1. What is your age range? (18-21 / 22-25 / 26-30 / 30+)

**Ans.** 26

2. How comfortable are you with using web apps on a scale of 1-5? (1 = not comfortable, 5 = very comfortable)

**Ans.** 5

3. Have you used a habit tracking app before? (Yes / No)

**Ans.** Yes

4. If yes, which one? (optional)

**Ans.** Cult Fit

5. How often do you try to build or track personal habits? (Never / Occasionally / Regularly)

**Ans.** Regularly

Detailed notes:

- **Notes for initial approach:** App is used to track habits someone's been doing
- **T1 (Create a habit):** Cleanly Executed
- **T2 (Log a completion):** Cleanly Executed
- **T3 (Find streak information):** Cleanly Executed
- **T4 (Set a weekly goal):** Cleanly Executed
- **T5 (Delete a habit):** Cleanly Executed

Post-test questionnaire results (Likert scales):

- **T1 (Create a habit): -**

Creating the habit was straightforward: 5

I knew where to go to create a habit: 5

- **T2 (Log a completion): -**

Logging a completion was easy: 5

I understood how to add a note to a log: 5

- **T3 (Find streak information): -**

Finding my streak information was easy: 5



The Analytics page was easy to navigate: 5

- **T4 (Set a weekly goal): -**

Setting a weekly goal was intuitive: 5

I understood where the goal setting feature was: 5

- **T5 (Delete a habit): -**

Deleting a habit was easy: 5

I felt confident the habit and its data would be removed: 5

- **Overall**

The app was easy to use overall: 5

I would use this app to track my own habits: 5

Any comments or suggestions for improvement?

[Straightforward, user friendly, easy to navigate]

## **Prioritized list of issues and corresponding changes:**

Issue: Participant 2 noted the add note feature during log completion has scope to improve

Change: Add sample common notes to click and fill easily

Priority: Could - High (Must, Could, Should, Would)

Was it implemented? How?

[Yes - quick-fill buttons added below the notes input showing common responses like "Felt great!", "Completed ✓", "Tough but done"]

Issue: Participant 1 noted that date range filter for 'this year' was missing

Change: Add a filter by 'this year'

Priority: Should - Medium

Was it implemented? How?

[Yes - "This Year" preset was added to the Analytics date range filter, along with Today, Past Week, and Past Month presets for more flexible filtering]

Issue: Participant 1 noted 0 values appeared flat in analytics with no visual distinction

Change: Show an empty state or placeholder when a habit has no recent activity or keep as is

Priority: Would - Low

Was it implemented? How?

[Not implemented - 0 values accurately reflect no logged activity. Issue was extremely specific to one particular user.]

Issue: Participant 1 noted two analytics KPI tiles appeared visually similar in color

Change: Review color differentiation across analytics cards

Priority: Would - Low

Was it implemented? How?

[Not implemented - color choices are intentional and consistent with the app's color palette. Each KPI card uses a distinct accent color (blue, teal, orange, green). Extremely user specific issue]

**Note:** Minor user-specific issues such as personal color preferences and individual feature requests were not actioned as they did not represent patterns across multiple participants and would not improve the experience for the broader target audience.

# HabitForge Usability Study Report

- Report date: 2026-04-18
- Authors: Aniket Nandi, Runze (Neil) Wang
- Study period: 2026-04-17 to 2026-04-18
- Application under test: HabitForge ( <https://habit-forge-fvl4.onrender.com> )
- Source transcripts: `usability-test-transcript.md` (xu), `usability-test-transcript-p2.md` (alex), `usability-test-transcript-p3.md` (parker)
- Session recordings (YouTube, unlisted):
  - xu (P1): <https://youtu.be/uT6Bg1hNWQg>
  - alex (P2): <https://youtu.be/otwUt0qwWbQ>
  - parker (P3): [https://youtu.be/E\\_Yo1EWkGsl](https://youtu.be/E_Yo1EWkGsl)

## Executive summary

Three participants completed moderated Zoom sessions covering the three core HabitForge workflows: creating a habit, logging and reviewing completions, and analyzing progress with date-range filters and sorting. All three completed every task without moderator assistance. The overall impression across participants was positive, and every issue uncovered during the study was minor - none prevented any task from being completed and none reduced any Likert score below 3 / 5.

The study surfaced one correctness item and four minor UX items that appeared in two or three independent sessions:

- Minor correctness item: the current-streak calculation may count non-consecutive completion days as consecutive. Observed by parker, needs a small backend verification.
- Minor UX item: streak and last-logged date are not visible on the habit card. All three participants asked for this.
- Minor UX item: the weekly completion percentage column is not explained in the UI. All three participants had to derive or guess the formula.
- Minor UX item: the log list is sorted oldest-first; two of three participants expected newest-first.
- Minor UX item: the “weekly” frequency is not defined in the UI (calendar week vs rolling seven days). Raised by alex, implicit in parker’s math reasoning.

Three further minor items were surfaced by single participants: lowercase category labels on the card, the absence of a password recovery flow, and a “description” field label that discourages use. One aspirational feature, an “at risk” streak indicator, emerged from parker’s coaching

perspective.

Participants’ overall effectiveness scores averaged 4.0 / 5. Overall intuitiveness averaged 3.67 / 5. The gap between these two scores, consistent across all three participants, points at HabitForge being functionally complete but slightly under-explained at the UI level - a tractable fix set.

## Note on email-dependent features

Several small items in this report (password recovery, “Continue with Google” single-sign-on, email verification at registration) depend on an outbound email capability. HabitForge is a class project and does not provision a commercial transactional email service (SendGrid, Postmark, AWS SES, etc.), since those services require a paid plan to send mail from a verified domain. Features that would require sending email are therefore intentionally out of scope for this iteration and are documented in the issue list with that rationale, rather than treated as omissions.

## Methodology

- Format: 1:1 moderated Zoom sessions with audio, video, and participant screen share recorded to the cloud.
- Session length: 30-minute slots, actual working duration between 23:12 and 26:14 - all sessions exceeded the 15-minute minimum.
- Moderators: Neil (xu, parker), Aniket (alex).
- Structure per session: welcome and consent (~3 min), demographics (~1 min), initial-approach / intuitiveness walkthrough (~2 min), T1 create habit, T2 log and read streak, T3 analytics with date range and sort, post-test Likert questionnaire, open feedback, wrap-up.
- Scoring: 1-5 Likert per task on two axes (effective, intuitive) plus overall effectiveness and overall intuitiveness.
- Success criteria per task were pre-defined in `usability-test.md`.

## Participants

| ID | Name   | Persona                     | Age | Background                                           | Web comfort (1-5) |
|----|--------|-----------------------------|-----|------------------------------------------------------|-------------------|
| P1 | xu     | Consistency-seeking student | 27  | CS student, last year. Tried Notion, gave up.        | 5                 |
| P2 | alex   | Returning self-improver     | 32  | Ops manager. Previously kept a paper bullet journal. | 4                 |
| P3 | parker | Coach-minded planner        | 28  | Master’s in information science, HCI TA.             | 5                 |

All three participants completed all three tasks without the moderator unblocking them. None needed the fallback test account. No technical failures (crashes, errored requests, broken links) were observed in any session.

## Task-by-task findings

### T1 - Create a habit

Success criterion: create “Read 20 pages” in the study category, weekly x4, under 3 minutes, unassisted.

| Participant | Time to complete | Assisted | T1 effective | T1 intuitive |
|-------------|------------------|----------|--------------|--------------|
| xu          | 1:40             | No       | 5            | 5            |
| alex        | 2:05             | No       | 5            | 4            |
| parker      | 2:40             | No       | 5            | 4            |

All three participants finished well under the 3-minute target. Likert scores for effectiveness were unanimously 5. The slight deductions on intuitiveness came from:

- alex: “week” is not defined anywhere (Monday-Sunday vs rolling seven days), and category labels render in lowercase on the habit card, looking unpolished.
- parker: the description field is labeled “optional”, which discourages filling in the “why does this habit matter to me” prompt that a coach would recommend capturing.

Positive side-finding from parker: toggling the Daily / Weekly radio resets the weekly-count field rather than carrying a stale value forward. This was noted as a good default.

### T2 - Log completions and read streak

Success criterion: log today and “two days ago”, correctly report current streak and most recent completion, under 4 minutes.

| Participant | Time to complete | Assisted | T2 effective | T2 intuitive |
|-------------|------------------|----------|--------------|--------------|
| xu          | 4:10             | No       | 4            | 3            |
| alex        | 4:30             | No       | 4            | 3            |
| parker      | 5:35             | No       | 3            | 3            |

All three participants went over the 4-minute target. The extra time was spent hunting for the current streak, which is only surfaced on the Analytics page. The habit card on the Dashboard shows name, category, frequency, and the View Logs / Edit / Delete controls, but no streak and no last-logged date. All three participants expected at least one of these on the card, and each

explicitly asked for it in open feedback.

Additional findings tied to this task:

- Two of three participants (alex, parker) noticed that the log list is sorted oldest-first and felt this was backwards. alex framed it in terms of her bullet journal: “newest on top” matched her mental model. parker framed it as “the thing you just did is the thing you want to see first”.
- parker uncovered a suspected streak-calculation bug. After logging only April 16 and April 18 (with a gap on April 17), the Analytics view reported a current streak of 2. Backfilling April 17 then produced a current streak of 3, which is internally consistent only if the streak is being computed as “distinct completion days in a recent window” rather than “the length of the current consecutive run”. A streak of 2 is not accurate when there is a missed day in between. This is flagged as a correctness issue (Priority: Must) rather than a UX one and should be verified in the analytics route before any cosmetic work on streaks.

### T3 - Analytics with date range and sort

Success criterion: set the view to the last 14 days, sort by best-performing, correctly identify best and worst habits, under 4 minutes.

| Participant | Time to complete | Assisted | T3 effective | T3 intuitive |
|-------------|------------------|----------|--------------|--------------|
| xu          | 3:40             | No       | 4            | 3            |
| alex        | 3:50             | No       | 4            | 3            |
| parker      | 4:10             | No       | 4            | 3            |

All three participants correctly used the From / To date pickers and the Sort by Weekly Completion % control. All three identified the best and worst performing habits. All three had trouble reproducing the weekly completion percentage numerically in their heads and requested an inline explanation of the formula.

- xu proposed a tooltip.
- alex proposed a tooltip plus a raw count (“5 of 12”).
- parker reverse-engineered the formula mid-task as “completions / target over range” and recommended a tooltip stating that explicitly, plus a separate info icon on the streak metric that defines what counts as a consecutive day.

parker’s unique contribution on this task was a feature idea: an “at risk” indicator for habits whose streak is about to break today. This came from his Sunday review ritual - “the question I always ask is what did you nearly miss”.

### Post-test Likert summary across all participants

| Metric | xu | alex | parker | Mean |
|--------|----|------|--------|------|
|--------|----|------|--------|------|

| Metric                             | xu | alex | parker | Mean |
|------------------------------------|----|------|--------|------|
| T1 effective (create habit)        | 5  | 5    | 5      | 5.00 |
| T1 intuitive (create habit)        | 5  | 4    | 4      | 4.33 |
| T2 effective (log + review)        | 4  | 4    | 3      | 3.67 |
| T2 intuitive (log + review)        | 3  | 3    | 3      | 3.00 |
| T3 effective (analytics + filters) | 4  | 4    | 4      | 4.00 |
| T3 intuitive (analytics + filters) | 3  | 3    | 3      | 3.00 |
| Overall effective                  | 4  | 4    | 4      | 4.00 |
| Overall intuitive                  | 4  | 4    | 3      | 3.67 |

Effectiveness consistently outranked intuitiveness by 0.33 to 0.67 points per category. This pattern matches the qualitative observation that HabitForge’s underlying features work, but the UI does not fully explain itself.

## Positive findings

- Landing and register flows were frictionless for all three participants. Each found the primary call-to-action in under 15 seconds and completed registration without assistance.
- The Dashboard empty state (“No habits yet” + a large “+ New Habit” button) successfully directed each participant to the correct next action.
- Habit creation is fast and low-friction. Mean time 2:08, every participant scored effectiveness a 5.
- Form state management is careful. parker explicitly praised the Daily / Weekly toggle for resetting the weekly-count rather than carrying over stale state.
- The Analytics date-range filter and sort controls were discoverable. Every participant used them correctly on the first attempt.
- Two of three participants (xu and alex) volunteered unprompted that they would consider using HabitForge as a real tool. alex specifically praised the absence of gamification.

## Prioritized list of issues and corresponding changes

All findings were minor. None blocked task completion and none drove any Likert score below 3 / 5. From the full set of observations (see the task-by-task findings section and the three transcripts for the complete list), the three most important items are prioritized below using the Must / Should / Could / Would scheme.

- Issue: No “Continue with Google” single-sign-on and no password recovery flow. xu asked unprompted for Google sign-in during the intuitiveness phase (“I don’t love making a new username and password for everything”); parker flagged the absence of password recovery

as something that would stop him from recommending HabitForge to mentees. Change: Add Google OAuth to the Register / Login flow and a standard email-based password reset, with a clear “Forgot password” link on the Login page. In the interim, add a short disclosure on the Register page: “Accounts are identified by username only. Password recovery is not currently supported - please store your password securely.”

- Priority: Would for the OAuth and full recovery flow (deferred pending budget); Should for the interim disclosure line.
- Was it implemented? How? Not implemented. Both capabilities depend on outbound transactional email - OAuth for the verified-email hand-off from Google, password reset for the reset-link email itself - which in turn requires a commercial email provider such as SendGrid, Postmark, Mailgun, or AWS SES. Those services only offer sending from a verified domain on paid plans, and HabitForge is a non-commercial class project with no budget for ongoing service fees. The team chose to ship a self-contained username / password flow rather than add a recurring paid dependency or a half-working OAuth stub. These features can be revisited if the project moves to a hosted commercial setting that funds an email service.
- Issue: Participants suggested the fixed category list feels limiting and asked for either more built-in categories or the ability to add their own. Raised lightly across sessions - alex noted that “Write morning pages” did not fit cleanly into study, and parker mentioned that coaching use cases often need cross-cutting categories like “relationships” or “side-project”. Change: Either expand the built-in category list or replace the dropdown with a create-on-the-fly combo box that stores user-defined categories per account. Would need a small schema addition so categories are scoped per user rather than hard-coded, and a migration for existing habits.
  - Priority: Would. Nice-to-have, not blocking.
  - Was it implemented? How? Not implemented. For the current app, the built-in categories (study, fitness, coding, health, other) cover every habit created during the three usability sessions and are considered sufficient for the submitted scope. Custom categories are a reasonable future enhancement once there is real multi-user traffic, since the value of user-defined tags grows with the breadth of habits being tracked. Deferred until that signal exists.